

Mrs Cooper
Mrs Greenberg
Mrs Kerr
Mrs Millar
Mrs Squires

Mr Feng
Ms Hoang
Ms Lau
Ms Peng
Mrs Ward

Name:

Teacher:



Pymble Ladies' College

YEAR 11

MATHEMATICS ADVANCED

ASSESSMENT TASK 1

**Term 1
2023**

General Instructions

- Working time 55 minutes
- Write using black or blue pen.
- Erasable pens are not to be used.
- Approved calculators may be used.
- Diagrams are not drawn to scale.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or untidy work.
- Reference sheets are provided.

Mark	/42
Rank	/
Highest Mark	/42

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Mathematics Advanced

Name:

Teacher's Name:

Section I – Multiple Choice Answer Sheet

Select the alternative (A), (B), (C) or (D) that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

Assessment Task 1

1 A ☐ B ☐ C ☐ D ☐

2 A ☐ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☐ D ☐

4 A ☐ B ☐ C ☐ D ☐

5 A ☐ B ☐ C ☐ D ☐

6 A ☐ B ☐ C ☐ D ☐

7 A ☐ B ☐ C ☐ D ☐

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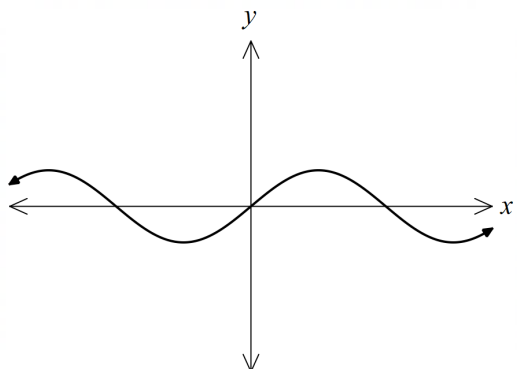
SECTION I

(7 Marks)

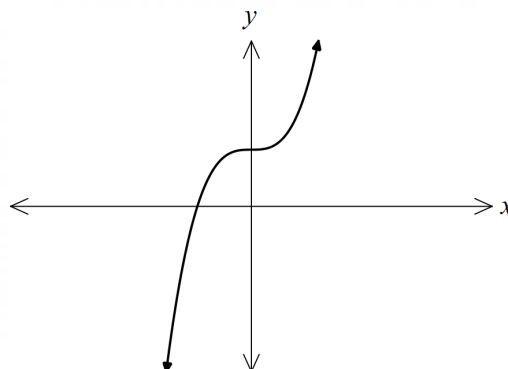
Answer using the Multiple Choice Answer Sheet provided.

1 Which graph shows the graph of an odd function?

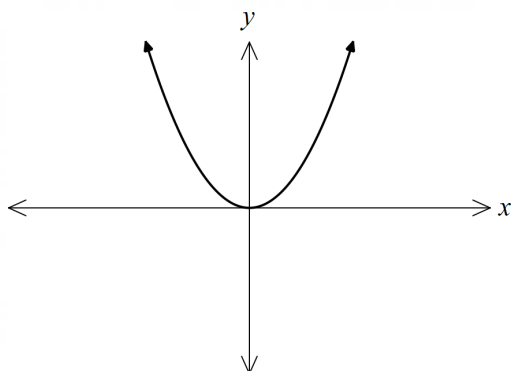
(A)



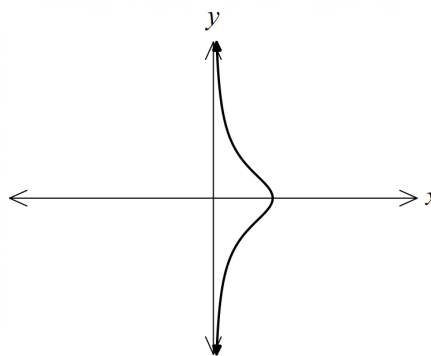
(B)



(C)



(D)



2 Which inequality defines the domain of the function $f(x) = \frac{1}{\sqrt{x+3}}$?

(A) $x > -3$

(B) $x \geq -3$

(C) $x < -3$

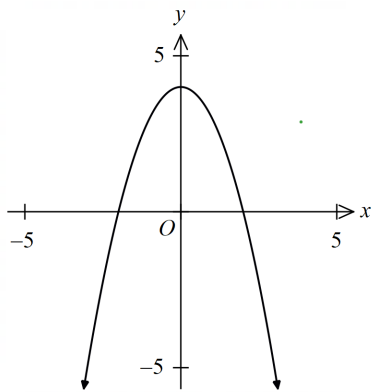
(D) $x \leq -3$

3 Which of the following is true for the equation $3x^2 - x - 2 = 0$?

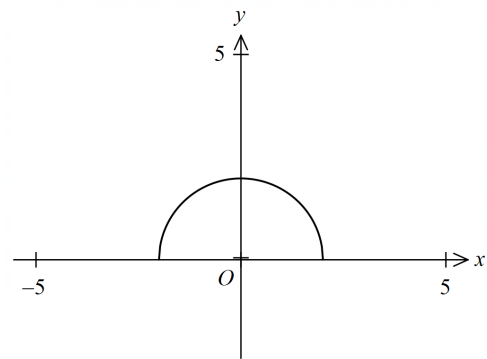
- (A) No real roots
- (B) One real root
- (C) Two real distinct roots that are rational
- (D) Two real distinct roots that are irrational

4 Which of the following functions is one-to-one?

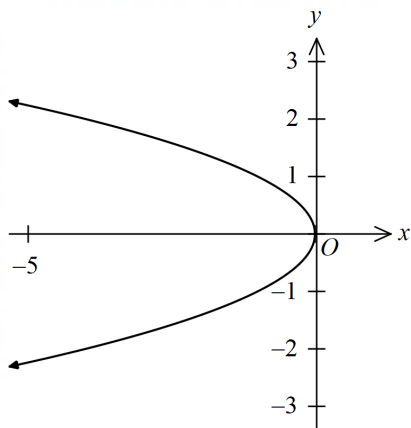
(A)



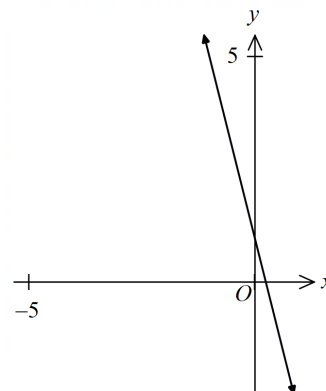
(B)



(C)



(D)



5 Which of the following is NOT equivalent to $\sqrt{24}$?

(A) $2\sqrt{6}$

(B) $\frac{1}{2}\sqrt{48}$

(C) $\sqrt{12} \times \sqrt{2}$

(D) $2\sqrt{3} \times \sqrt{2}$

6 The function $f(x) = \frac{x^2}{1-x^3}$ is which of the following:

(A) even

(B) odd

(C) neither

(D) cannot be determined

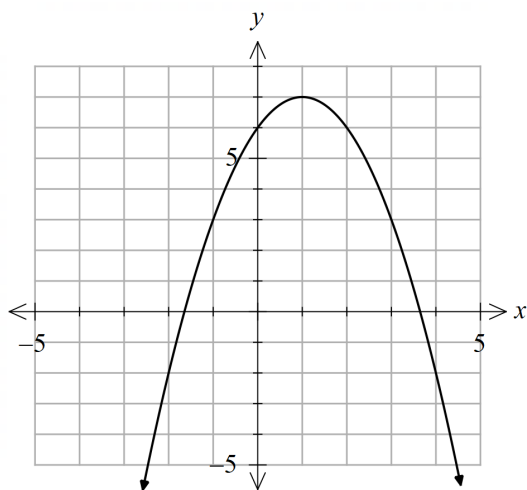
7 For the graph of the quadratic function below, what is the inequality that satisfies $f(x) > 3$?

(A) $(-\infty, -1] \cup [3, \infty)$

(B) $(-\infty, -1) \cup (3, \infty)$

(C) $[-1, 3]$

(D) $(-1, 3)$



End of Section I

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SECTION II

(35 Marks)

Attempt all questions

Answer these questions in the spaces provided

Your responses should include relevant mathematical reasoning and/or calculations.

Question 8 (4 marks)**Marks**

(a) Solve for m $\frac{a^5 \sqrt{a}}{a^{-2}} = a^m$

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(b) Express $y = 2x^2 - 12x + 11$ in the form $y = a(x - h)^2 + k$.

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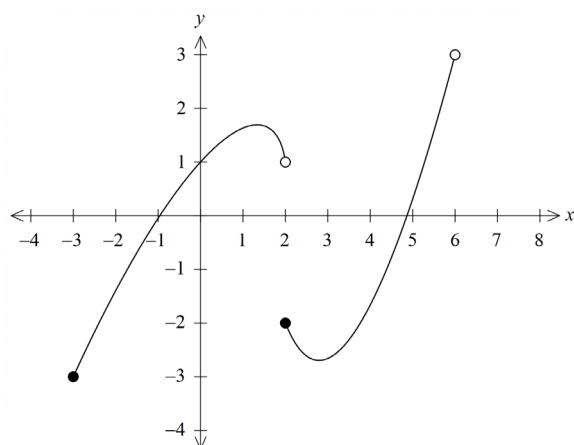
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Question 9 (4 marks)**Marks**

- (a) State the domain and range of the following functions.

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- (b) $y = 1 - \sqrt{1 - x}$

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Question 10 (6 marks)**Marks**

- (a) A function is defined as $f(x) = x^3 - 2x$.

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Show this function is either even, odd or neither.

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- (b) The function $y = f(x)$ is defined as follows:

2

$$f(x) = \begin{cases} x^2 & \text{for } x \leq 2 \\ 2x + 1, & \text{for } x > 2 \end{cases}$$

Evaluate $f(-1) + 2f(2) - f(3)$.

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- (c) If $g(x) = x^2 + 3x + 5$, show that $g(a+1) = g(a) + 2(a+2)$.

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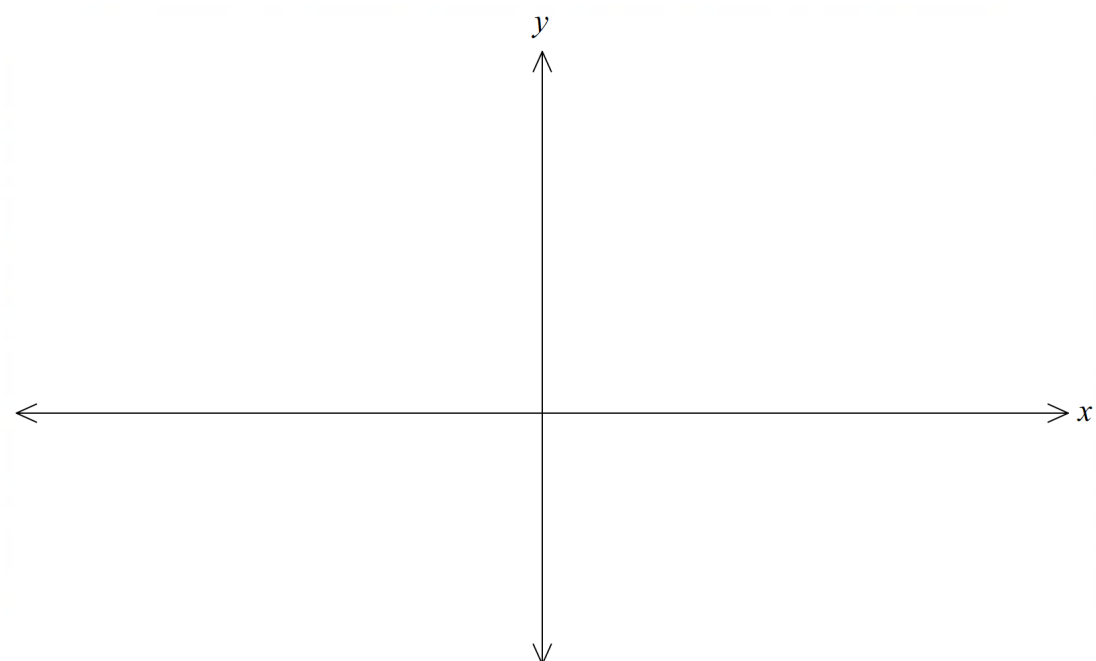
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Question 11 (6 marks)**Marks**

- (a) (i) On the axis provided, sketch the graph $y = x(x-4)^2$, showing all intercepts.

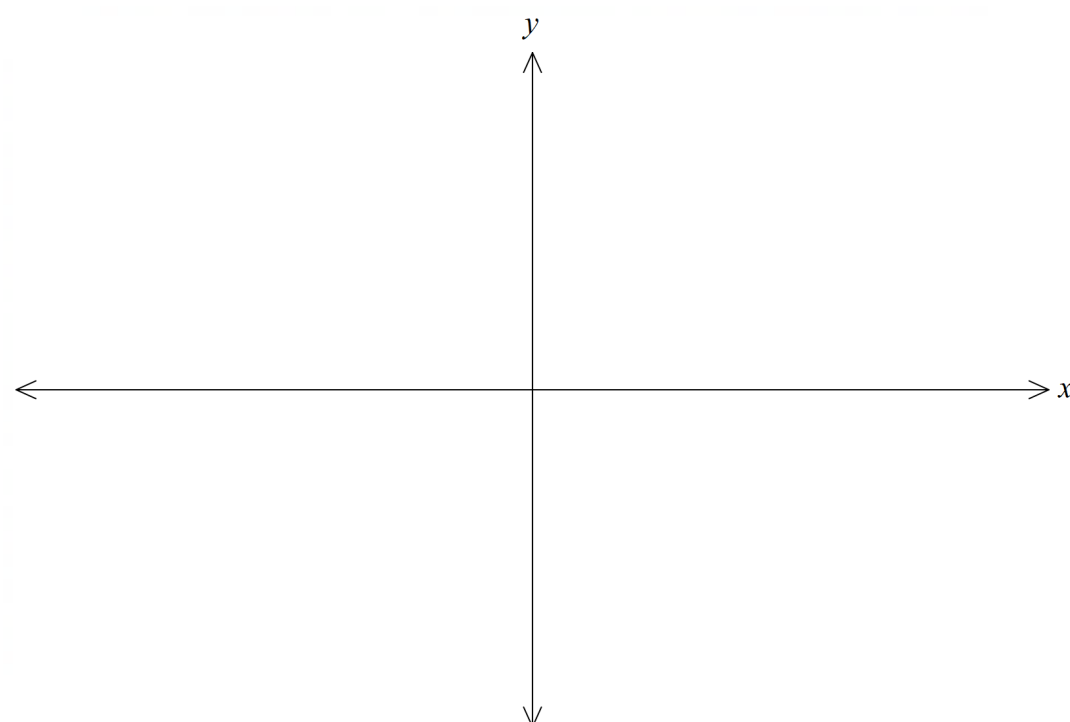
2

- (ii) Hence, find the values of x for which $x(x-4)^2 > 0$.

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- (iii) On the number plane below, sketch $y = f(-x)$ labelling all intercepts.

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Question 12 (7 marks)**Marks**

- (a) Given that $9x^3 + 3x^2 - 12x - 4 = (3x^2 - 4)(3x + 1)$, hence, or otherwise, solve $18x^4 + 6x^3 - 24x^2 - 8x = 0$.

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- (b) For what values of m does the equation $6x^2 - 12mx - 7m + 3 = 0$ have equal roots.

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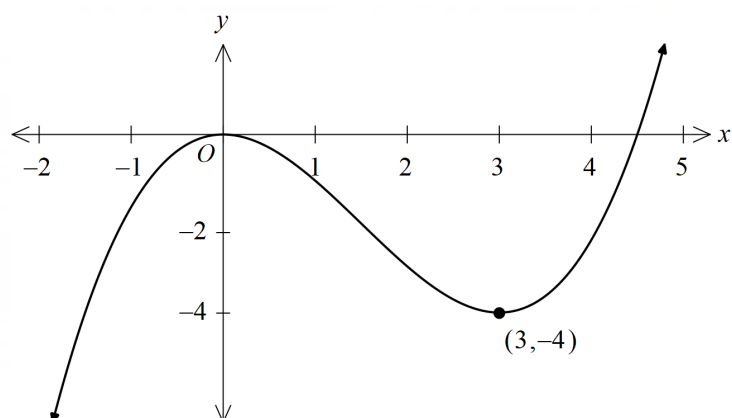
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Question 12 continues on next page

Question 12 continued.

(c) The graph of $y = f(x)$ is sketched below.



For what values of k does the equation $f(x) = k$ have one solution?

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Question 13 (4 marks)

Marks

Let $f(x) = -x^2 + x + 4$ and $g(x) = x^2 - 2$.

(a) Find $g(f(3))$.

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(b) Express $f(g(x))$ in the form $ax^4 + bx^2 + c$, where a , b and c are non-zero integers.

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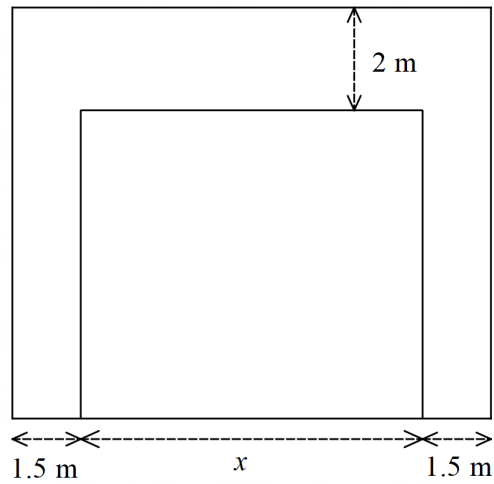
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Question 14 (4 marks)

Marks

A square garage door of length x metres is surrounded on three sides, by brickwork. The brick work is 1.5 metres wide on each side of the door and has a height of 2 metres above the door, as shown below.



NOT TO SCALE

- (a) Find an expression for the area of the brickwork. 2

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- (b) To meet building regulations, the area of the brick wall must be exactly twice the area of the garage door. 2

Find the value(s) of x that satisfy the building regulations.

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Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

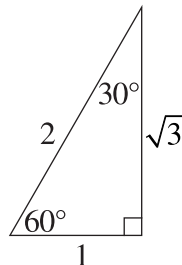
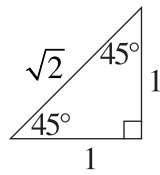
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

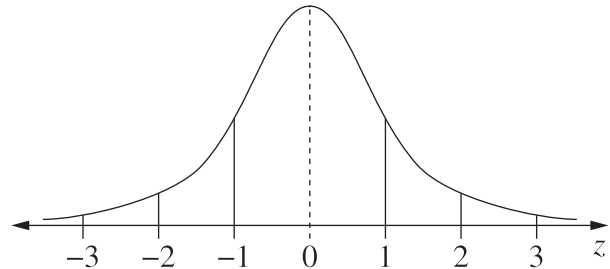
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \cdots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}_nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

Mathematics

Advanced

Name:

Teacher's Name:

Section I – Multiple Choice Answer Sheet

Select the alternative (A), (B), (C) or (D) that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct
↑

Assessment Task 1

1 A ☒ B ☐ C ☐ D ☐

2 A ☒ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☒ D ☐

4 A ☐ B ☐ C ☐ D ☒

5 A ☐ B ☒ C ☐ D ☐

6 A ☐ B ☐ C ☒ D ☒

7 A ☐ B ☐ C ☐ D ☒

SECTION II

(35 Marks)

Attempt all questions

Answer these questions in the spaces provided

Your responses should include relevant mathematical reasoning and/or calculations.

Question 8 (4 marks)

Marks

(a) Solve for m $\frac{a^5 \sqrt{a}}{a^{-2}} = a^m$

2

$$\frac{a^{11/2}}{a^{-2}} = a^{15/2}$$

$$= a^{11/2 - (-2)} \quad \therefore m = 15/2$$

2	Correct
1	error

(b) Express $y = 2x^2 - 12x + 11$ in the form $y = a(x-h)^2 + k$.

2

$$y = 2(x^2 - 6x + 9) + 11 - 18$$

$$= 2(x-3)^2 - 7$$

2	Correct
1	Correctly completing the square

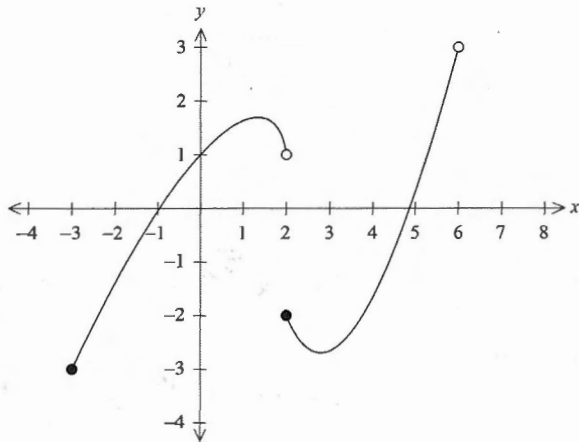
having $(x-3)^2$

Question 9 (4 marks)

Marks

(a) State the domain and range of the following functions.

2



① max if no $x \in$
 $y \in$
and both D and R
correct

Domain : $x \in [-3, 6)$	2	Both correct
Range : $y \in [-3, 3)$	1	one correct

(b) $y = 1 - \sqrt{1-x}$

2

Domain : $1-x \geq 0 \quad x \leq 1$	2	Both correct
Range : $y \leq 1$	1	one correct

① max if no $x \in$
 $y \in$
and both D and R
correct.

Question 10 (6 marks)

Marks

- (a) A function is defined as $f(x) = x^3 - 2x$.

2

Show this function is either even, odd or neither.

$$\begin{aligned}
 f(a) &= a^3 - 2a \\
 f(-a) &= (-a)^3 - 2(-a) \\
 &= -a^3 + 2a \\
 &= -(a^3 - 2a) \\
 &= -f(a) \\
 \therefore \text{Function is odd}
 \end{aligned}$$

2	Correct
1	Subs $x = -a$ or equivalent

- (b) The function $y = f(x)$ is defined as follows:

2

$$f(x) = \begin{cases} x^2 & \text{for } x \leq 2 \\ 2x+1 & \text{for } x > 2 \end{cases}$$

Evaluate $f(-1) + 2f(2) - f(3)$.

$$\begin{aligned}
 &= (-1)^2 + 2(2)^2 - (2(3)+1) \\
 &= 1 + 8 - 7 \\
 &= 2
 \end{aligned}$$

2	Correct
1	error

① if correct equation
eg; $x^2 + 2x^2 - (2x+1)$

- (c) If $g(x) = x^2 + 3x + 5$, show that $g(a+1) = g(a) + 2(a+2)$.

2

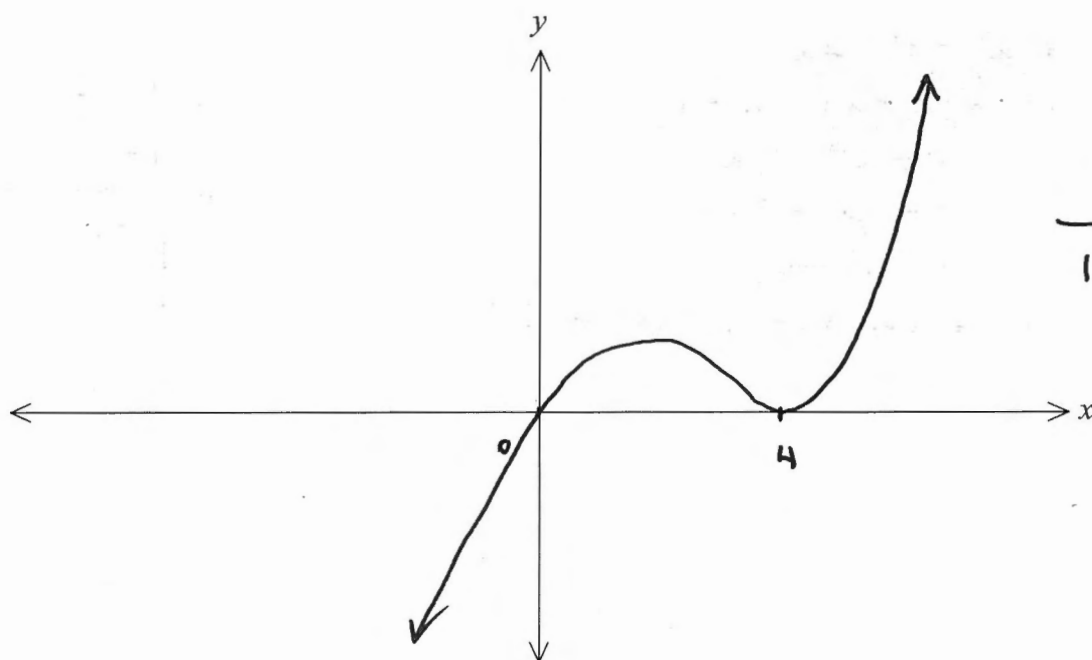
$$\begin{aligned}
 g(a+1) &= (a+1)^2 + 3(a+1) + 5 \\
 &= a^2 + 2a + 1 + 3a + 3 + 5 \\
 &= a^2 + 5a + 9 \\
 g(a) &= a^2 + 3a + 5 \\
 g(a) + 2(a+2) &= a^2 + 3a + 5 + 2a + 4 \\
 &= a^2 + 5a + 9 \\
 &= g(a+1) \text{ As req'd}
 \end{aligned}$$

2	Correct
1	Substituting $x = a+1$ correctly

Question 11 (6 marks)

Marks

- (a) (i) On the axis provided, sketch the graph $y = x(x-4)^2$, showing all intercepts. 2



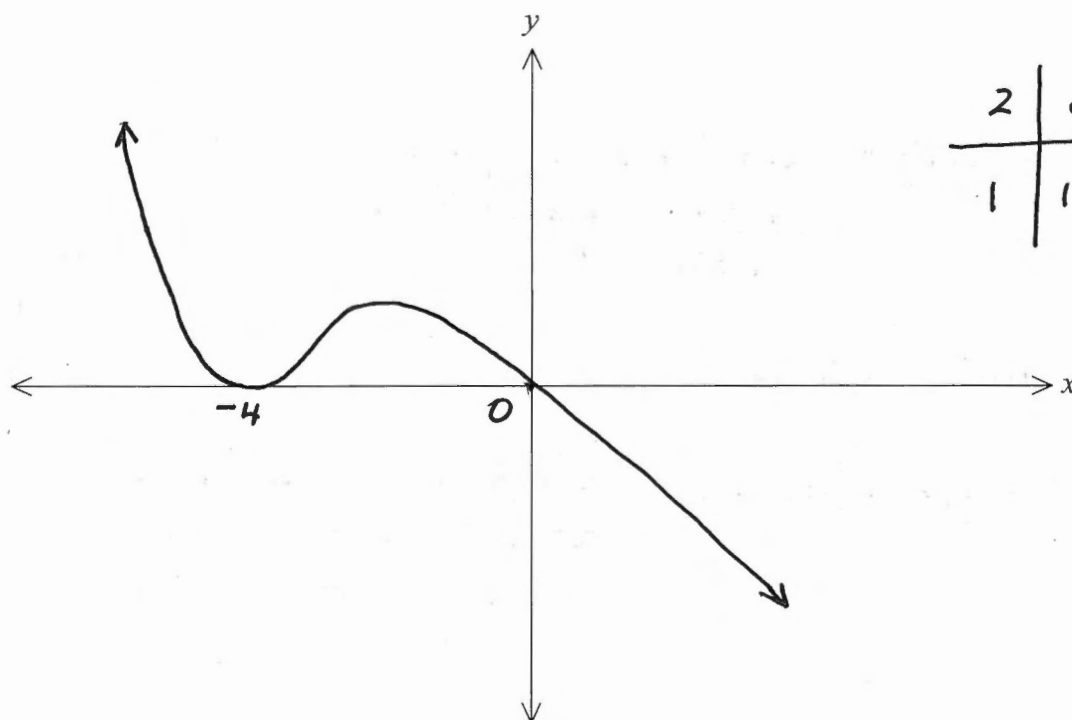
2	correct
1	error

- (ii) Hence, find the values of x for which $x(x-4)^2 > 0$. 2

$x \in (0, 4) \cup (4, \infty)$

2	correct
1	error

- (iii) On the number plane below, sketch $y = f(-x)$ labelling all intercepts. 2



2	correct
1	error

Question 12 (7 marks)

Marks

- (a) Given that $9x^3 + 3x^2 - 12x - 4 = (3x^2 - 4)(3x + 1)$, hence, or otherwise, solve $18x^4 + 6x^3 - 24x^2 - 8x = 0$.

2

$$2x(9x^3 + 3x^2 - 12x - 4) = 0$$

$$2x(3x^2 - 4)(3x + 1) = 0 \quad *$$

$$\therefore x = 0 \text{ or } x^2 = \frac{4}{3} \text{ or } x = -\frac{1}{3}$$

$$x = \pm \frac{2}{\sqrt{3}}$$

2	Correct
1	Correct ✗ factorisation

- (b) For what values of m does the equation $6x^2 - 12mx - 7m + 3 = 0$ have equal roots.

3

$$\Delta = (-12m)^2 - 4(6)(3 - 7m)$$

$$= 144m^2 - 72 + 168m$$

$$= 24(6m^2 + 7m - 3)$$

3	Correct
2	Correct Δ error in factorisation
1	$\Delta = 0$

Equal roots $\Delta = 0 \therefore 24(6m^2 + 7m - 3) = 0$

$$3m - 2 \quad 6m^2 + 7m - 3 = 0$$

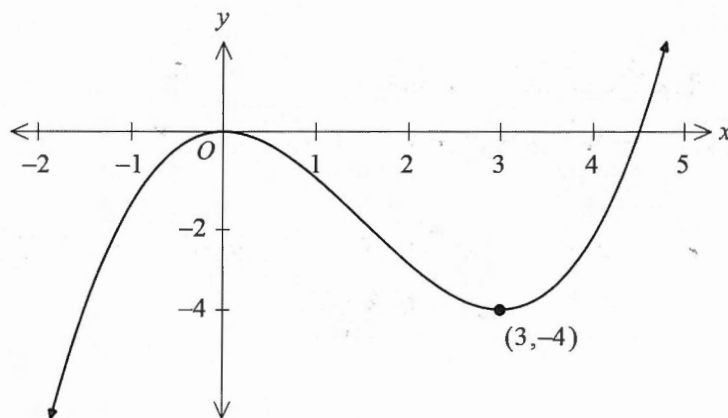
$$2m \times 3 \quad (3m - 2)(2m + 3) = 0$$

$$\therefore m = \frac{2}{3} \text{ or } m = -\frac{3}{2}$$

Question 12 continues on next page

Question 12 continued.

(c) The graph of $y = f(x)$ is sketched below.



For what values of k does the equation $f(x) = k$ have one solution?

2

$k > 0$ or $k < -4$ 2 | correct

 1 | one correct

Question 13 (4 marks)

Marks

Let $f(x) = -x^2 + x + 4$ and $g(x) = x^2 - 2$.

(a) Find $g(f(3))$.

2

$$\begin{aligned} g(-3^2 + 3 + 4) &= g(-2) \\ &= (-2)^2 - 2 \\ &= 2 \end{aligned}$$

2	correct
1	error

(b) Express $f(g(x))$ in the form $ax^4 + bx^2 + c$, where a , b and c are non-zero integers.

2

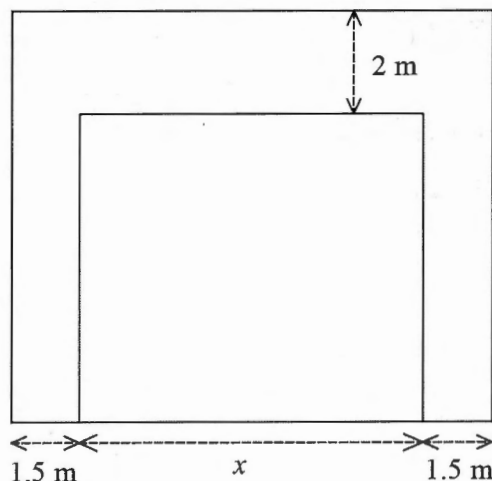
$$\begin{aligned} f(x^2 - 2) &= -(x^2 - 2)^2 + (x^2 - 2) + 4 \\ &= -(x^4 - 4x^2 + 4) + x^2 + 2 \\ &= -x^4 + 5x^2 - 2 \end{aligned}$$

2	correct
1	Correct to *

Question 14 (4 marks)

Marks

A square garage door of length x metres is surrounded on three sides, by brickwork. The brick work is 1.5 metres wide on each side of the door and has a height of 2 metres above the door, as shown below.



NOT TO SCALE

- (a) Find an expression for the area of the brickwork.

2

$$\begin{aligned} A_{\text{Brickwork}} &= (x+3)(x+2) - x^2 \quad * \\ &= x^2 + 5x + 6 - x^2 \\ &= 5x + 6 \end{aligned}$$

2	Correct
1	Correct to *

- (b) To meet building regulations, the area of the brick wall must be exactly twice the area of the garage door.

2

Find the value(s) of x that satisfy the building regulations.

$$\begin{aligned} 5x + 6 &= 2x^2 \\ 2x^2 - 5x - 6 &= 0 \end{aligned}$$

$$x = \frac{5 \pm \sqrt{25 + 48}}{4}$$

$$x = \frac{5 \pm \sqrt{73}}{4}$$

2	Correct
1	Correct quadratic and use of quadratic formula, not eliminating negative result

But x is a length $\therefore x > 0 \therefore x = \frac{5 + \sqrt{73}}{4}$ m.

End of Assessment Task